

667 Studio

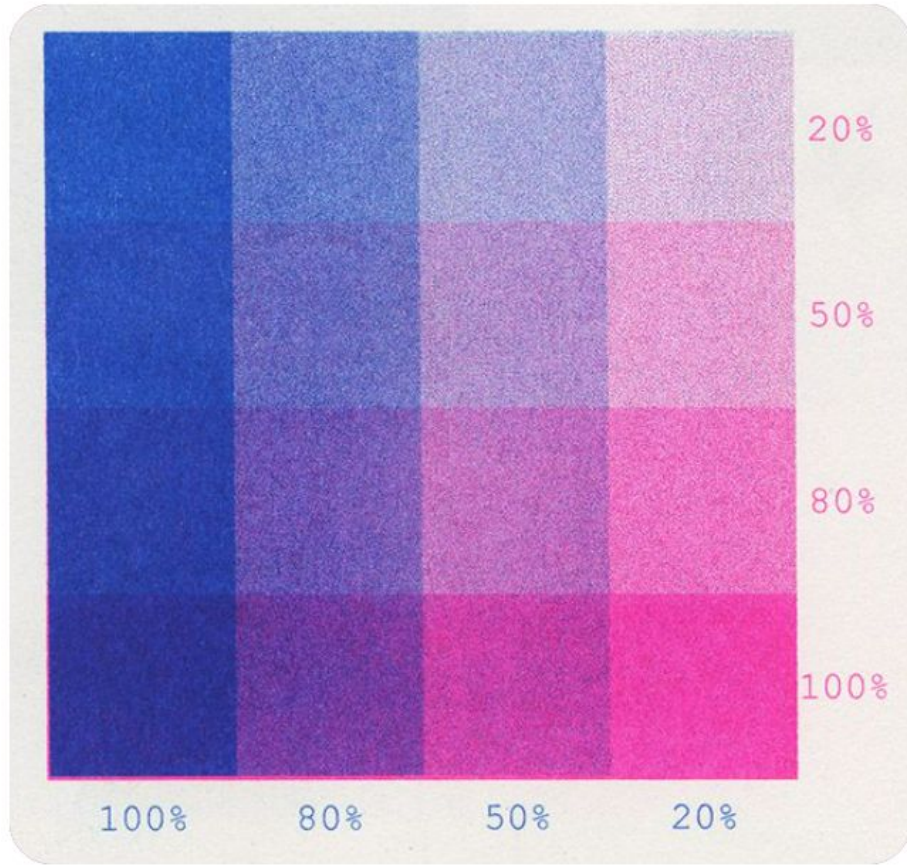
Risograph Preparation and Printing



Risograph printing is a technique best described as 'digital screen printing'. The process is similar to screen printing, but with the convenience of an office copier. It's known for its vivid colours (that other printers can't easily produce) and its signature textures. The printer uses inks rather than toner which allows for vivid, bright colours. These machines deliver 'perfect imperfections' and provide a cheap and easy method for reproducing work at large quantities, like posters, prints and zines.* It uses uncoated paper as the paper helps the ink to dry through absorption.

*Riso Pop - <https://www.risopop.com/what-is-riso>





With risograph printing, it's important to keep in mind that the inks are translucent, meaning you can layer them to make different colours. You can use different opacities with your ink layers to create different colours, tones and shades. This process is easiest to experiment with in programs like Photoshop, Illustrator or Procreate where you can adjust the opacity of your layers.

Risograph colours

Many risograph studios will list the available ink colours they have on their websites -

These swatches will often be accompanied by the official Riso colour name, sometimes a Pantone code, and the HEX code for the colour. The HEX code can be used to find the digital analogue of these colours in imaging programs.

These are the colours that are available for printing with in 667 Studio - Please note the symbols in the center of the swatches, as it denotes which machine has which colour! The 2 machines have different colour libraries with some overlap.

☆ means colour is exclusive to Doris (self service)

🌀 means colour available on both machines

△ means colour is exclusive to Gail (full service)



Black



Bright Red



Red



Fluo Pink



Fluo Orange



Sunflower



Yellow



Light Lime



Kelly Green



Teal



Hunter Green



Aqua



Metallic Gold



Blue



Medium Blue



Violet



Purple

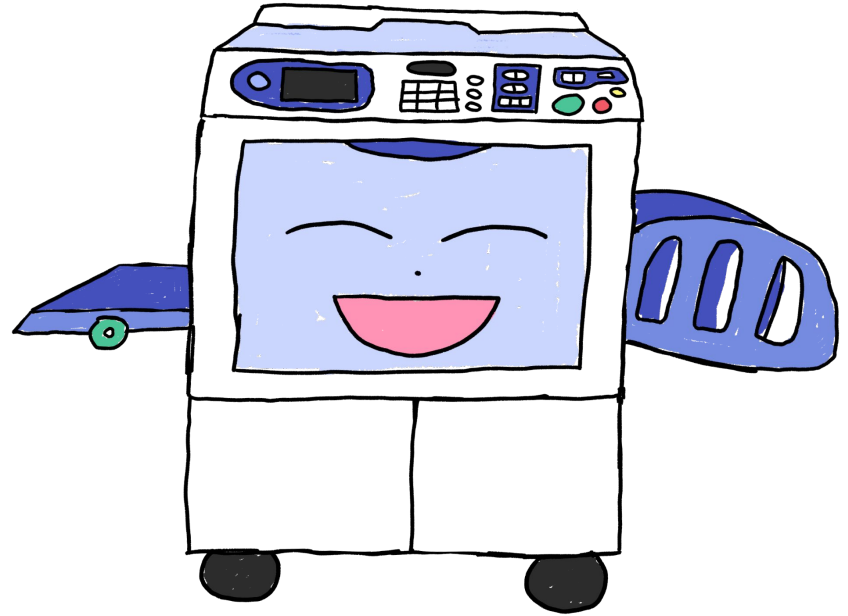
Doris (EZ591) and Gail (MH9450)

In 667 Studio, we have 2 risograph printers.

Doris is an EZ591 model. This is a single drum printer, meaning the machine prints with 1 colour at a time. This way of risograph printing will give you the most thorough understanding of the printmaking process. This machine is a self-service machine, meaning students print for themselves independently. We have a daily training session on this machine every Wednesday at 2:30pm. If this does not work for you, we can train you at a different time.

Gail is an MH9450 model. This is a dual drum printer, meaning the machine prints with 2 colours at the same time. This cuts down on the time it requires to do risograph printing. This machine is a full service machine, meaning the staff of 667 Studio do the printing for you. Please note, this method costs more and has longer turnaround times.

Please refer to our Canvas Course page for pricing and turnaround times!

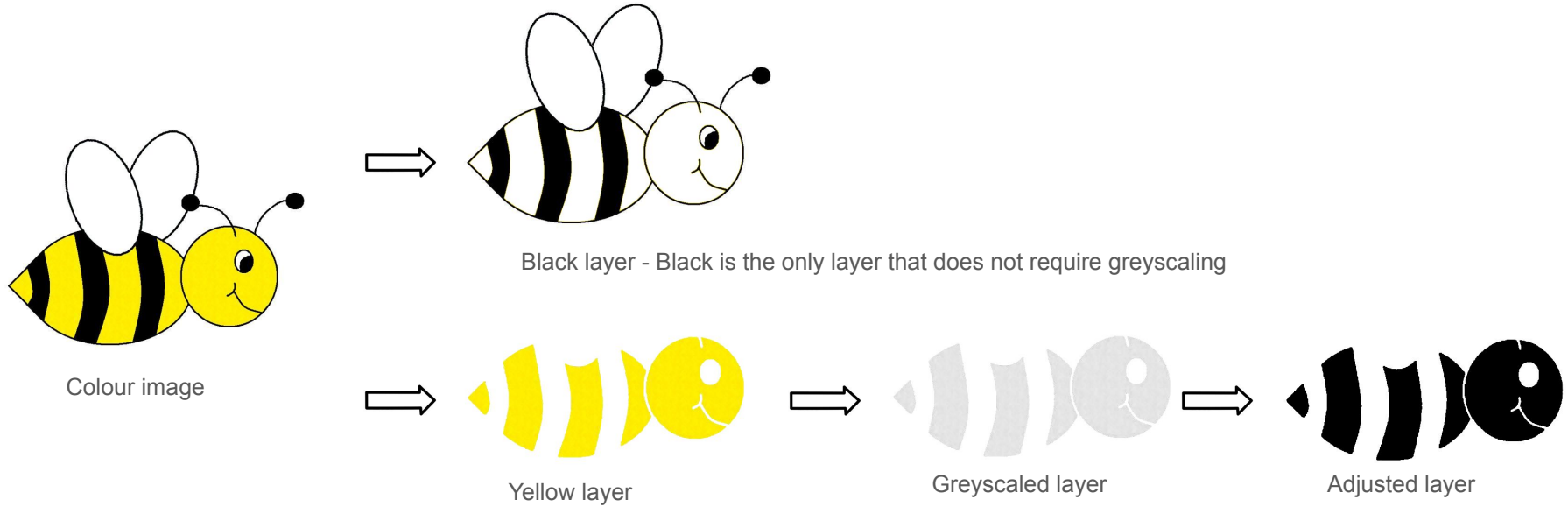


Risograph Basics



The risograph does not read colour files, instead it reads grayscale data from your artwork. The risograph reads 100% black in your file as 100% deposit of whichever ink colour you wish to assign to that layer. Lighter black tones will equate to a lighter colour of that assigned ink. As you can see here, we are printing this grayscale image with violet ink. Anywhere where there is black, that is where the ink will be at 100% opacity. The lighter purples correspond to the lighter black tones in the artwork.

Risograph Basics



When you are creating grayscale files of your layers, it is important to keep in mind how certain colours translate to grayscale. For example, if you have a yellow layer, yellow is very bright but lacks tone. When you greyscale it, it will turn to a very light gray. If you want a saturated yellow in your image, you'll need to adjust the levels or curves of the file in order to get your image to print properly. We have specific instructions on this process on our [Canvas Page](#).



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Here is an example of how our separate colour layers look by themselves, and how they will look once they are combined. This is a digital mock-up, but the effects that the risograph would produce are very similar. Computer screens cannot accurately show fluorescent colours though, so the real print will be a lot more vibrant in real life.

Gradient map tool for risograph experimentation

One of the most handy tools for digital risograph mock-up is the gradient map tool. To demonstrate that, here is a small ink drawing. Let's say I scanned this drawing into my computer and wanted to experiment how it will look printed in one of the risograph ink colours. Always consult the website of the print studio you are using to be sure the colours you want to use are available first!

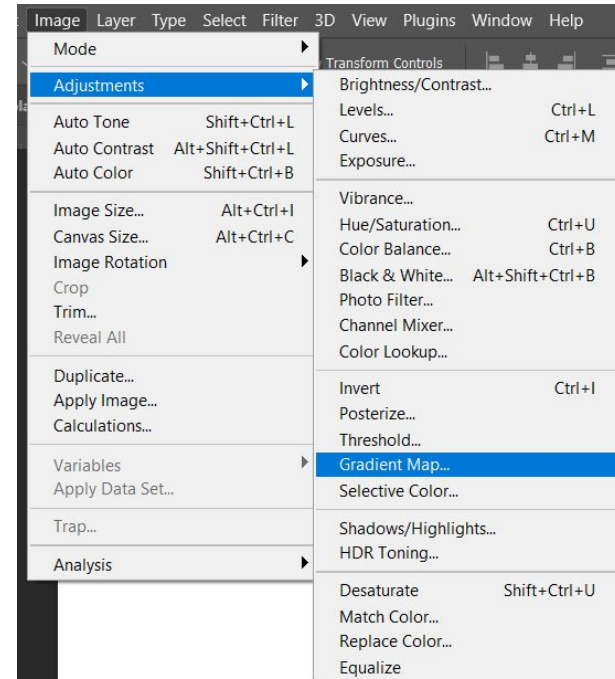
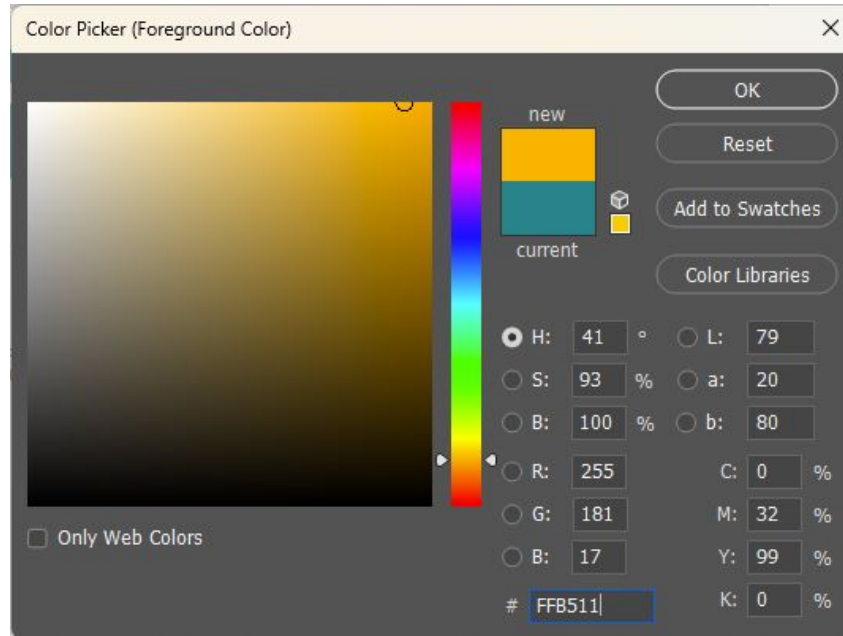


Art by Linnea Sterte

If I wanted to see how my image would print in sunflower, the easiest method is to use the gradient map tool. You can enter the HEX code of a colour into the HEX # field at the bottom of the Color Picker field, or you can use the eyedropper tool to sample them directly. Once I have my colour, I'm going to use the gradient map tool.



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And now, we have a mock-up of how our image is going to look! You can create more layers to add more colours to get an even more vibrant colour scheme.

The risograph doesn't just read pure black and white, but also gray tones. So it can replicate pencil, watercolour and other textures.



Artwork by Merna Schumann



GRAIN TOUCH



DOT PROCESS

The most popular texture of risograph printing is referred to as 'Grain Touch.' This is a process where gradients and tone in your image are duplicated through incredibly fine dots in the screen to create an image with a signature fuzzy texture. 'Dot Process' or 'Screen-covered' uses a halftone pattern for your printing, giving you a more graphic impression.

Step by step guides for Photoshop and Procreate

If you enroll in our Canvas Course, we have more PDFS that will give you specific step-by-step instructions on how to correctly format your files for risograph printing. Files for the risograph need to be grayscale before printing, so be sure to consult our resources to understand how to do this. You can also come by and ask any of our studio team, but we prefer you do some reading for yourself first!



Artwork by Gabby Golec

Formatting!

- **PAPER SIZE:** The paper size we stock in 667 is 11" x 17". This size is also called tabloid or ledger.
- **BORDERS:** The risograph cannot print full bleed on 11" x 17" paper. Make sure you have at least a border of .25" on each side to ensure your prints come out properly. To achieve full bleed, you will need to cut into the 11" x 17" paper.
- **FILE FORMAT:** Please send grayscale PDFs with the 2001 preset when you are submitting a file for full-service printing.
- **FILE NAMING:** Please name each of your files with the colour you wish to print with in the file name. Ex: Student_Yellow.pdf
- **INK COVERAGE:** Please note that heavy ink coverage can jam the machine. We recommend for full coverage areas, you change your opacity to 85%.



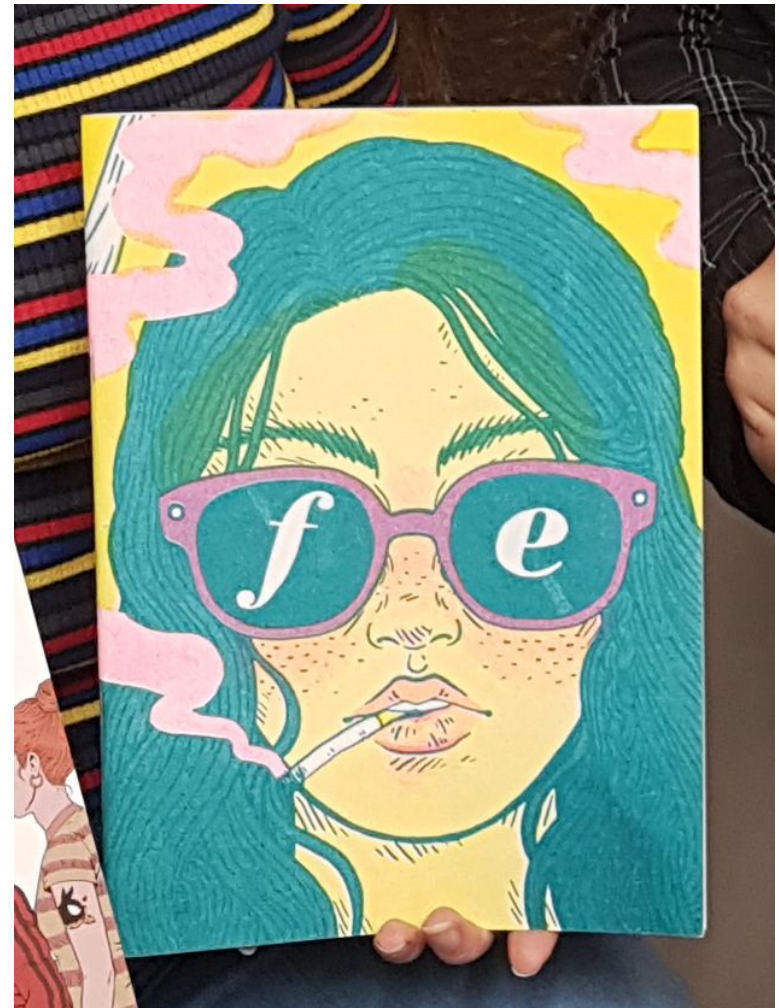
Things to watch out for!

- Make sure to double-check your layers on multiply to make sure you aren't creating any extra colours / dark areas you don't want in your final print. You may need to clean up under your linework or various colour layers to get a cleaner image
- Be sure to talk to your printer about what is the best way to prep your file - Different riso machines have different functions and that's the best way to be prepared
- Be sure to always be aware of how much of a gutter you need around your image! All printers generally leave 1/4" border on sheets of paper, so you will need to trim your papers down to get a full bleed

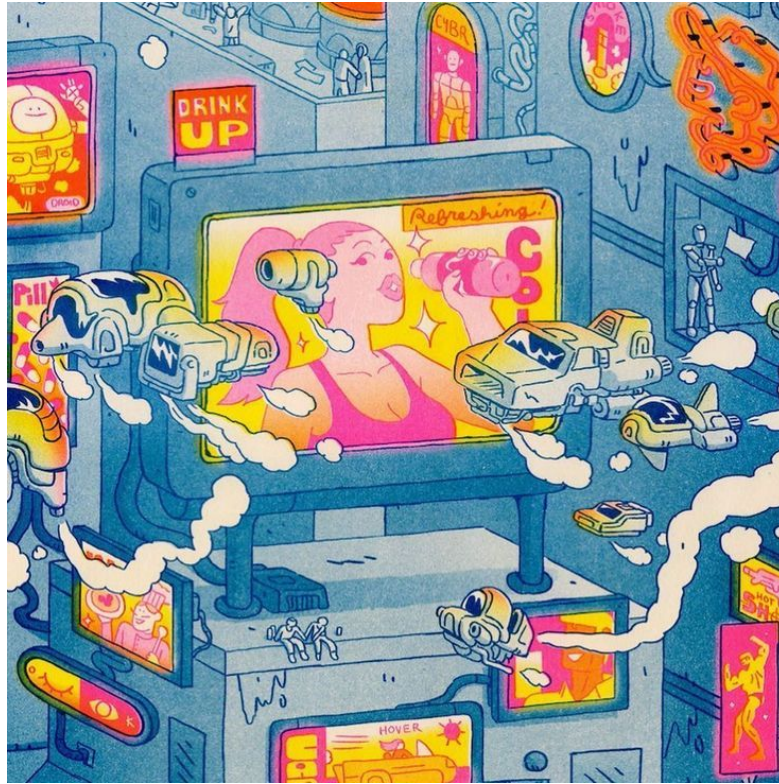


An example of mistakes that can occur if you don't clean up your layers -

If you look at the hair, you can see the front of the hair are more green than the teal colour. This was one of the first risograph projects I did and didn't realize that sitting the teal on top of the yellow would result in this colour mixing. Just one of the things to watch out for when you are planning your print.



Get inspired by these incredible risograph artists!



Wren McDonald!



Natalie Andrewson!



Jasmine Floyd!

Risograph Resources!

- Stencil Wiki - A wiki resource for all things Risograph, including a comprehensive colour list of available riso inks.

stencil.wiki | [A RISO wiki for artists, designers, and printers](#)

- Spectrolite - A MAC desktop app that assists in colour separation and experimentation

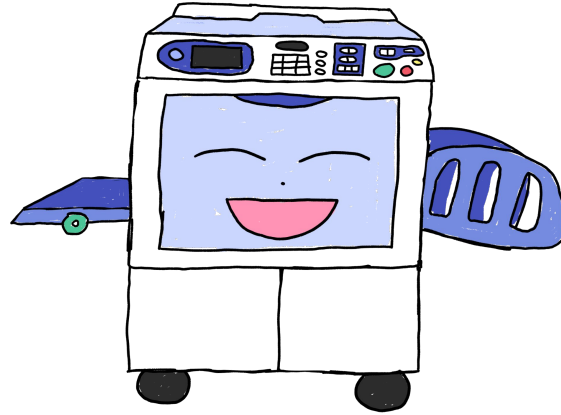
<https://spectrolite.app/>

Local Printers:

- VidePress, Toronto
- Colour Code Printing, Toronto
- Material Object, Toronto
- Paper Pusher, Toronto
- Bullywug Press, Toronto
- Pindot Press, Oakville



You can always contact us with questions!



You can e-mail us at 667@ocadu.ca for quotes, information or general questions. We also have a canvas course where you can download various slideshows and access more information. The QR code and URL is located on the door of the studio!